# DLR-TomoSAR — Training stability & regularization

leaky clamp · LR warmup · adaptive clipping · augmentation

July 2026

# Why clamp the output — the exponential overflows

physical — log space — normalized

$$p \xrightarrow{\log(1+\,\cdot\,)} u \xrightarrow{(\,\cdot\,-\,m)/s} y$$

normalize — dataset build, once: input is ground truth, *bounded*

$$p \xleftarrow{e^{(\cdot)}-1} z \xleftarrow{s\,\cdot\,+m} \hat{y}$$

denormalize — every prediction: input is $\hat{y}$, *any* real

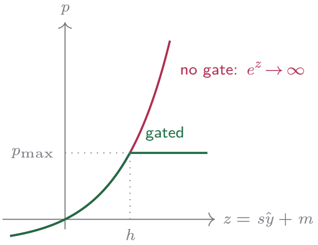
$p \in \{a, \sigma\}$ ($\mu$ skips the log — linear both ways); $m, s$: fixed channel shift & scale. Composed: $p = e^{s\hat{y}+m} - 1$.

$$\begin{aligned} \hat{y} \to \infty \;&\Longrightarrow\; z \to \infty \\ &\Longrightarrow\; p = e^{z} - 1 \to \infty \\ &\Longrightarrow\; \mathcal{L} \to \infty \\ &\Longrightarrow\; \nabla_\theta \mathcal{L} = \mathrm{NaN} \end{aligned}$$

$\mathcal{L}$: training loss, $\theta$: weights. One overflowing pixel $\Rightarrow$ every weight NaN $\Rightarrow$ **no** training.

$$\boxed{p \;=\; e^{\min(z,\,h)} - 1 \;\le\; p_{\max}}$$

$h = \log(1 + p_{\max})$, $p_{\max}$: physical ceiling. The front-end `clamp` toggle — off $\Rightarrow$ overflow returns.

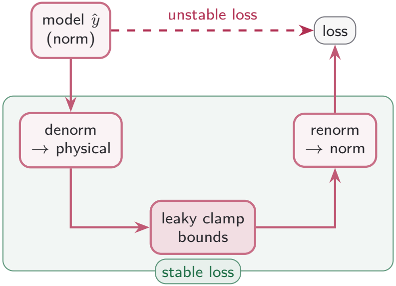



*unbounded $e^z$ (red) vs. exponent gated at $h$ (green)*

# The leaky clamp — stability keystone

One unphysical parameter (negative $\sigma$, runaway $a$, off-grid $\mu$) poisons every loss term downstream; a hard clamp fixes the forward pass but kills the gradient. The leaky clamp bounds the value *and* keeps a slope.

*forward pass — the clamp guards the loss path*

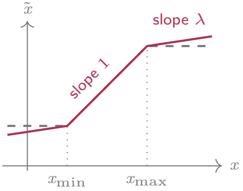



$$\boxed{\tilde{x} = \mathrm{clamp}(x) + \lambda\,(x - \mathrm{clamp}(x))}$$

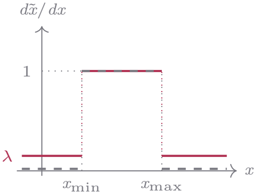
$$a \in [0,\, a_{\max}],\quad \mu \in [x_{\min},\, x_{\max}],\quad \sigma \in [\tfrac{1}{2}\,x_{\text{step}},\, \tfrac{1}{2}\,x_{\text{range}}]$$

*one outlier prediction $x$ (pushed off physical bounds)*

| on the outlier | loss | gradient $\partial\mathcal{L}/\partial x$ |
|---|---|---|
| no clamp | $\infty$ / NaN | poisons all terms |
| hard clamp | bounded | $0$ — slot dies |
| leaky clamp | bounded | $\lambda > 0$ — keeps signal |




value: training, leaky vs inference, hard ($\lambda = 0$)




gradient: never zero — saturated slots keep receiving signal

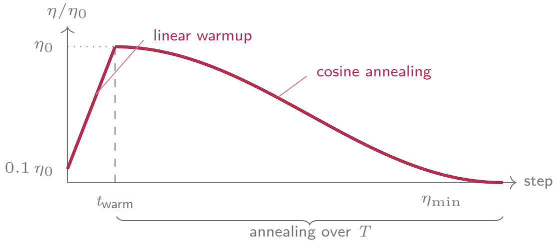
# LR warmup and adaptive gradient clipping

- A linear **warmup** over the first 200 steps tames the early, high-variance gradients, then epoch-level cosine annealing down to $\eta_{\min}$.

$$\eta(t, s) = \eta_0 \, F_{\cos}(t) \, f_{\text{warm}}(s)$$

$$f_{\text{warm}}(s) = 0.1 + 0.9 \, \min\big(1, \; s / t_{\text{warm}}\big)$$

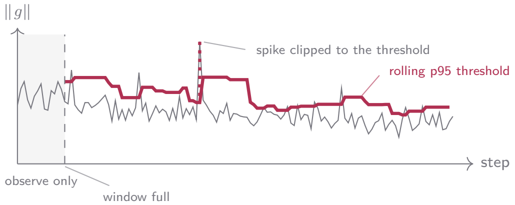
$$F_{\cos}(t) = \tfrac{1}{2}\big(1 + \cos(\pi(t - t_{\text{warm}})/T)\big)$$




- Rolling window of the last 200 global gradient norms, threshold recomputed every step as the p95 of the window — the first ∼200 steps only observe; this avoids starving the gradients — a training insurance.

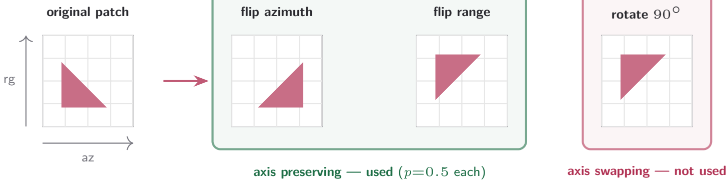
$$\tau_t = \mathrm{p95}\big(\{\|g\|\}_{t-200:t}\big)$$

$$g \leftarrow g \cdot \min\big(1, \; \tau_t / \|g\|\big)$$

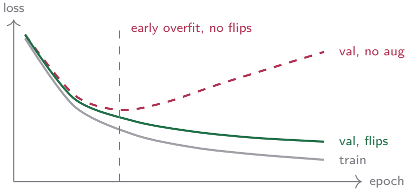


# Augmentation is the regularizer

One scene is the whole dataset, so the regularizer matters more than usual. The candidate transforms, screened by acquisition geometry:




- Flips are applied **jointly to input and target** ($p$=0.5 each): four orientations of every patch, spatial correspondence intact — and each axis keeps its physical identity.
- Rotation would exchange range and azimuth — two physically distinct axes (resolution, acquisition geometry, statistics) — so only the axis-preserving pair is used.




The flips push the validation turn-off point out — the single-scene overfit arrives much later.

# Augmentation, measured — flips delay the overfit and halve the seed spread

| metric | no aug | flips | Δ |
|---|---|---|---|
| *training (single scene, validation)* | | | |
| best epoch (val) | 26.2 | 43.8 | +17.6 |
| best val loss ↓ | 0.1443 ±.0008 | 0.1420 ±.0007 | −0.0023 |
| *reconstruction (curve)* | | | |
| curve $R^2$ ↑ | 0.573 ±.022 | 0.569 ±.008 | −0.004 |
| curve MAE ↓ | 0.0501 ±.0012 | 0.0496 ±.0008 | −0.0005 |
| PSNR [dB] ↑ | 50.36 ±.22 | 50.32 ±.08 | −0.04 |
| per-pixel $R^2$ ↑ | 0.359 ±.050 | 0.359 ±.031 | −0.001 |
| profile cos med ↑ | 0.851 ±.020 | 0.850 ±.024 | −0.001 |
| SSIM elev ↑ | 0.9582 ±.0009 | 0.9586 ±.0006 | +0.0004 |
| SSIM range ↑ | 0.9257 ±.0027 | 0.9258 ±.0021 | +0.0002 |
| SSIM azim ↑ | 0.9457 ±.0014 | 0.9461 ±.0012 | +0.0004 |
| peak err mean ↓ | **4.47** ±.86 | 4.70 ±.47 | +0.23 |
| median ↓ | 0.94 ±.37 | **0.81** ±.30 | −0.13 |
| p95 ↓ | 15.44 ±.67 | 15.44 ±.47 | +0.00 |
| *detection (matched)* | | | |
| precision ↑ | 0.816 ±.015 | 0.821 ±.008 | +0.005 |
| recall ↑ | 0.656 ±.056 | 0.641 ±.029 | −0.015 |
| matched F1 ↑ | 0.727 ±.040 | 0.720 ±.020 | −0.007 |
| count exact ↑ | 0.611 ±.066 | 0.595 ±.034 | −0.016 |
| *seed-to-seed spread (std over 5 seeds) ↓* | | | |
| recall | .056 | .029 | −48% |
| count exact | .066 | .034 | −49% |
| peak err mean | .86 | .47 | −46% |
| per-pixel $R^2$ | .050 | .031 | −38% |

**runs**: UNet · conv · sorted-GT · $K=2$ · param-MSE · presence none · full norm ladder · patch $64\times32$ (aug = column); cells: mean ± std over five seeds. Training rows: validation split; others: held-out test, $5.0$ M px. **bold** = better mean, marked only where the two differ by more than $5\%$; Δ = flips − no aug, coloured (gain / cost) only where $|\Delta|$ clears it; spread rows compare the seed stds; green values = the training shift and the spread halving — the effects that clear it. Peak median/p95: per-pixel peak-error distribution; cos med: median profile cosine; peak in elevation units.

# Augmentation by scatterer count — no accuracy shift, half the spread

| metric | no aug | flips | $\Delta$ |
|---|---|---|---|
| matched precision ↑ | 0.816 $\pm$.015 | 0.821 $\pm$.008 | $+0.005$ |
| $k$=1 | 0.848 $\pm$.019 | 0.853 $\pm$.008 | $+0.005$ |
| $k$=2 | 0.776 $\pm$.013 | 0.782 $\pm$.007 | $+0.006$ |
| matched recall ↑ | 0.656 $\pm$.056 | 0.641 $\pm$.029 | $-0.015$ |
| $k$=1 | 0.647 $\pm$.088 | 0.620 $\pm$.046 | $-0.028$ |
| $k$=2 | 0.670 $\pm$.016 | 0.672 $\pm$.008 | $+0.002$ |
| count acc $\mid$ pred $k$=1 ↑ | 0.952 $\pm$.009 | 0.948 $\pm$.007 | $-0.004$ |
| count acc $\mid$ pred $k$=2 ↑ | 0.720 $\pm$.013 | 0.733 $\pm$.010 | $+0.013$ |
| matched $\mu$ MAE ↓ | 2.30 $\pm$.12 | 2.30 $\pm$.08 | $+0.00$ |
| $k$=1 | 0.63 $\pm$.06 | **0.59** $\pm$.04 | $-0.04$ |
| $k$=2 | 4.07 $\pm$.05 | 4.05 $\pm$.08 | $-0.02$ |
| matched $\sigma$ MAE ↓ | 0.86 $\pm$.05 | 0.87 $\pm$.03 | $+0.01$ |
| $k$=1 | 0.25 $\pm$.02 | 0.25 $\pm$.01 | $+0.00$ |
| $k$=2 | 1.51 $\pm$.01 | 1.51 $\pm$.01 | $+0.00$ |
| matched $a$ MAE ↓ | 0.364 $\pm$.034 | 0.374 $\pm$.026 | $+0.010$ |
| $k$=1 | 0.155 $\pm$.029 | 0.161 $\pm$.017 | $+0.006$ |
| $k$=2 | 0.586 $\pm$.035 | 0.593 $\pm$.022 | $+0.007$ |

**runs**: UNet · conv · sorted-GT · $K$=2 · param-MSE · presence none · full norm ladder · patch $64 \times 32$ (aug = column); held-out test band, cells mean $\pm$ seed std over five seeded replicas; $k$ = true scatterer count of the pixel. **bold** = better mean, marked only where the two differ by more than 5%; $\Delta$ = flips − no aug, coloured only where $|\Delta|$ clears it; green $\pm$ = the spread halving — the one effect that does. Precision rewards under-prediction — rank on recall. count acc $\mid$ pred $k$: share of pixels claiming exactly $k$ scatterers whose true count is $k$. $\mu$, $\sigma$ in elevation units; $a$ in linear amplitude.

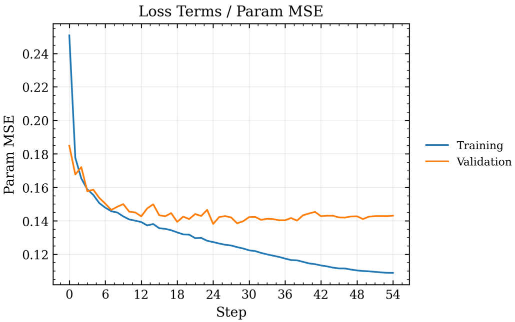
# Augmentation, predicted statistics vs the GT — same calibration, pinned down

| prediction mean | GT | no aug | flips | $\Delta$ |
|---|---|---|---|---|
| *slot 0 — strongest scatterer; GT-active on every pixel* | | | | |
| active frac | 1.00 | 0.706 $\pm$.066 | 0.685 $\pm$.035 | $-0.021$ |
| $a$ — active | 0.77 | 0.89 $\pm$.07 | 0.90 $\pm$.03 | $+0.01$ |
| $\mu$ — active | $-4.89$ | $-4.03$ $\pm$.15 | $-3.92$ $\pm$.10 | $+0.11$ |
| $\sigma$ — active | 1.76 | 1.52 $\pm$.08 | 1.48 $\pm$.05 | $-0.04$ |
| *slot 1 — second scatterer; GT-active on 26% of pixels* | | | | |
| active frac | 0.260 | 0.306 $\pm$.008 | 0.299 $\pm$.006 | $-0.007$ |
| $a$ — active | 1.43 | 1.06 $\pm$.05 | 1.04 $\pm$.03 | $-0.02$ |
| $a$ — inact. | 0.00 | 0.45 $\pm$.04 | **0.42** $\pm$.01 | $-0.03$ |
| $\mu$ — active | 11.34 | 9.97 $\pm$.40 | 10.09 $\pm$.33 | $+0.12$ |
| $\mu$ — inact. | — | 3.39 $\pm$.57 | 4.09 $\pm$.31 | $+0.70$ |
| $\sigma$ — active | 3.53 | 3.02 $\pm$.04 | 2.94 $\pm$.05 | $-0.08$ |
| $\sigma$ — inact. | — | 1.90 $\pm$.04 | 1.87 $\pm$.03 | $-0.03$ |
| *distribution width — std over GT-active pixels* | | | | |
| $a$ — slot 0 | 1.48 | 1.17 $\pm$.06 | 1.17 $\pm$.03 | $+0.01$ |
| $\mu$ — slot 0 | 3.30 | 2.03 $\pm$.12 | 1.99 $\pm$.07 | $-0.04$ |
| $\sigma$ — slot 0 | 2.84 | **1.01** $\pm$.03 | 0.95 $\pm$.03 | $-0.06$ |
| $a$ — slot 1 | 1.45 | 0.84 $\pm$.05 | 0.83 $\pm$.04 | $-0.01$ |
| $\mu$ — slot 1 | 16.24 | 8.76 $\pm$.27 | 8.72 $\pm$.18 | $-0.04$ |
| $\sigma$ — slot 1 | 4.01 | **1.14** $\pm$.04 | 1.07 $\pm$.02 | $-0.07$ |

**runs**: UNet · conv · sorted-GT · $K{=}2$ · param-MSE · presence none · full norm ladder · patch $64\times32$ (aug = column); held-out test band, cells mean $\pm$ seed std over five seeded replicas; means conditioned on the slot's GT activity mask. **bold** = closest to the GT column, marked only where best and worst differ by more than $5\%$; $\Delta$ = flips − no aug, coloured only where $|\Delta|$ clears it (cost = away from the GT); green $\pm$ = the seed spreads flips pin down. Inactive GT slots define no $\mu/\sigma$ reference (—). Width rows: std of the prediction over GT-active pixels vs the GT's own std. $\mu$, $\sigma$ in elevation units; $a$ in linear amplitude.

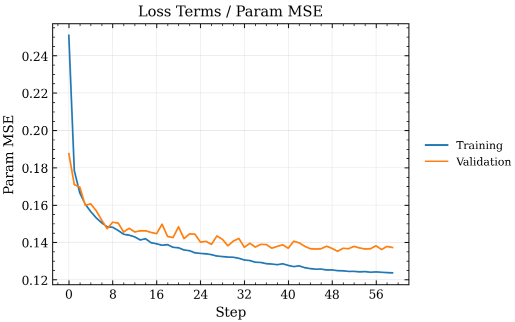
# Augmentation — the two loss curves

**no augmentation**




validation flattens early — training pulls away, the single-scene overfit sets in.

**flips ($p$=0.5 each)**




validation keeps descending — the train–val gap stays narrow for longer.